

TAKUA PA, Thailand

WMO: 485610

Lat: 8.6842N

Lon: 98.2522E

Elev: 8

StdP: 101.23

Time Zone: 7.00 (E07)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 10	(b) 21.5	(c) 22.3	(d) 18.0	(e) 12.9	(f) 26.6	(g) 19.1	(h) 13.9	(i) 26.3	(j) 5.4	(k) 26.7	(l) 4.8	(m) 27.0	(n) 0.0	(o) N/A	(p) 0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 7.6	(c) 34.2	(d) 27.1	(e) 33.5	(f) 26.8	(g) 32.9	(h) 26.6	(i) 28.4	(j) 32.8	(k) 28.0	(l) 32.2	(m) 27.6	(n) 31.6	(o) 1.1	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 27.2	(b) 23.0	(c) 31.3	(d) 26.8	(e) 22.4	(f) 30.7	(g) 26.5	(h) 22.0	(i) 30.2	(j) 91.5	(k) 33.0	(l) 89.6	(m) 32.3	(n) 87.9	(o) 31.8	(p) 31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.1	4.3	3.8	20.0	35.5	0.9	1.0	19.4	36.2	18.8	36.8	18.3	37.3	17.7	38.0	
			18.5	28.9	1.2	0.9	17.6	29.5	16.8	30.0	16.2	30.5	15.3	31.1	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	27.2	27.5	28.1	28.7	28.3	28.0	27.8	27.6	27.2	27.1	27.4	27.2
	DBStd	1.13	0.85	0.84	1.04	1.01	1.15	1.17	1.16	1.26	1.00	0.89	0.73	0.88
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6446	532	489	562	562	569	539	551	546	515	529	521	532
	CDD18.3	3404	273	256	303	312	311	289	293	287	265	271	271	274
	CDH23.3	35136	2721	2724	3384	3568	3450	3138	3118	2962	2493	2417	2531	2632
	CDH26.7	12766	1037	1144	1482	1528	1291	1107	1047	939	706	693	857	936
Wind	WSAvg	0.7	0.7	0.6	0.6	0.5	0.7	0.9	1.1	1.0	0.8	0.5	0.4	0.7
Precipitation	PrecAvg	3674	49	41	140	208	404	425	441	534	609	496	241	68
	PrecMax	5490	218	125	569	544	810	816	899	1006	1057	1035	598	240
	PrecMin	2528	0	2	7	17	170	146	205	230	288	223	87	12
	PrecStd	564	52	31	119	111	149	148	128	166	198	176	130	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.5	34.3	35.1	35.4	34.9	33.5	33.2	33.1	32.5	32.3	32.7	33.0
		MCWB	24.6	25.7	27.0	28.2	28.1	27.8	27.9	28.3	27.4	26.5	26.3	25.1
	2%	DB	32.6	33.3	33.9	34.3	33.3	32.4	32.1	31.9	31.6	31.4	32.0	32.2
		MCWB	24.9	25.5	26.7	27.7	27.7	27.4	27.2	27.3	26.9	26.4	26.1	25.3
	5%	DB	31.8	32.4	33.0	33.2	32.4	31.7	31.3	31.1	30.8	30.8	31.4	31.5
		MCWB	24.9	25.3	26.3	27.3	27.4	27.0	26.8	26.7	26.5	26.3	26.0	25.3
	10%	DB	31.0	31.6	32.3	32.4	31.6	31.0	30.6	30.5	30.1	30.1	30.8	30.8
		MCWB	24.8	25.1	26.1	27.0	27.1	26.8	26.6	26.5	26.3	26.2	25.9	25.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.7	27.2	28.3	28.8	28.7	28.5	28.6	28.6	28.0	27.6	27.3	27.1
		MCDB	31.3	32.4	33.5	34.3	33.3	32.4	32.5	32.5	31.7	31.1	31.3	31.1
	2%	WB	26.2	26.6	27.4	28.2	28.1	27.9	27.7	27.9	27.2	27.0	26.9	26.6
		MCDB	30.6	31.5	32.5	33.2	32.3	31.5	31.3	31.4	30.6	30.3	30.7	30.4
	5%	WB	25.9	26.1	27.0	27.7	27.7	27.4	27.2	27.2	26.8	26.6	26.5	26.1
		MCDB	30.1	30.8	31.8	32.4	31.6	30.9	30.4	30.3	29.9	29.8	30.2	29.9
	10%	WB	25.5	25.7	26.5	27.3	27.3	27.0	26.8	26.7	26.5	26.3	26.2	25.7
		MCDB	29.5	30.2	31.1	31.6	30.9	30.3	29.7	29.5	29.4	29.3	29.7	29.4
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.2	8.6	7.6	6.3	5.7	5.4	5.3	5.6	6.0	7.1	7.7
		MCDBR	9.5	10.4	9.9	9.0	7.6	6.8	6.6	6.3	6.8	7.2	8.1	8.7
		MCWBR	3.7	4.1	3.9	3.5	3.2	2.9	2.8	2.7	2.9	3.1	3.3	3.5
	5% WB	MCDBR	8.5	9.3	8.8	8.2	7.0	6.3	6.0	5.9	6.2	6.7	7.6	7.9
		MCWBR	3.7	4.0	3.7	3.5	3.2	3.0	2.9	2.9	3.0	3.1	3.4	3.5
Clear-Sky Solar Irradiance	taub	0.463	0.484	0.499	0.482	0.457	0.461	0.463	0.453	0.436	0.444	0.453	0.463	
	taud	2.293	2.211	2.211	2.264	2.379	2.386	2.361	2.385	2.427	2.397	2.356	2.306	
	Ebn at Noon	850	844	833	836	840	826	828	849	874	869	853	841	
	Edh at Noon	133	148	150	141	122	120	124	123	119	122	124	129	
All-Sky Solar Radiation	RadAvg	5.22	6.04	6.09	5.89	4.83	4.65	4.58	4.68	4.64	4.56	4.66	4.62	
	RadStd	0.38	0.34	0.54	0.37	0.38	0.34	0.30	0.33	0.35	0.46	0.34	0.40	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) +0.60	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (11 neighbors)	+0.16	+0.31	N/A	N/A	+0.18	+0.21	N/A	N/A	N/A	N/A

Nomenclature: See separate page